

Seven years later, Markowitz published his first book, *Portfolio Selection: Efficient Diversification of Investments* (John Wiley & Sons, 1959). In what many believe was his greatest contribution, he now turned his attention to measuring the riskiness of an entire portfolio. He called it *covariance*, a method for measuring the direction of a group of stocks. The more they move in the same direction, the greater is the chance that economic shifts will drive them down at the same time. By the same token, a portfolio composed of risky stocks might actually be a conservative selection if the individual stock prices move differently. Either way, Markowitz said, diversification is the key. The smart course for investors, he concluded, is first to identify the level of risk they are comfortable handling, and then to construct an efficient diversified portfolio of low-covariance stocks.

Eugene Fama—The Efficient Market

In 1965, Eugene Fama of the University of Chicago published his PhD dissertation, “The Behavior of Stock Market Prices,” in the *Journal of Business*, proposing a comprehensive theory of the behavior of the stock market. His message was very clear: Predictions about future stock prices are pointless because the market is too efficient. In an efficient market, as information becomes available, a great many smart people aggressively apply that information in a way that causes prices to adjust instantaneously, before anyone can profit. At any given point, prices reflect all available information and hence we say the market is efficient.

Bill Sharpe—Capital Asset Pricing Model

About 10 years after Markowitz’s paper first appeared, a young PhD student named Bill Sharpe talked with him at length about his work in portfolio theory and the need for countless covariances. The next year, in 1963, Sharpe’s dissertation was published: “A Simplified Model of Portfolio Analysis.” While fully acknowledging his reliance on Markowitz’s ideas, Sharpe suggested a simpler method. Sharpe believed that all securities bore a common relationship with some underlying base factor, and therefore analysis was simply a matter of measuring the